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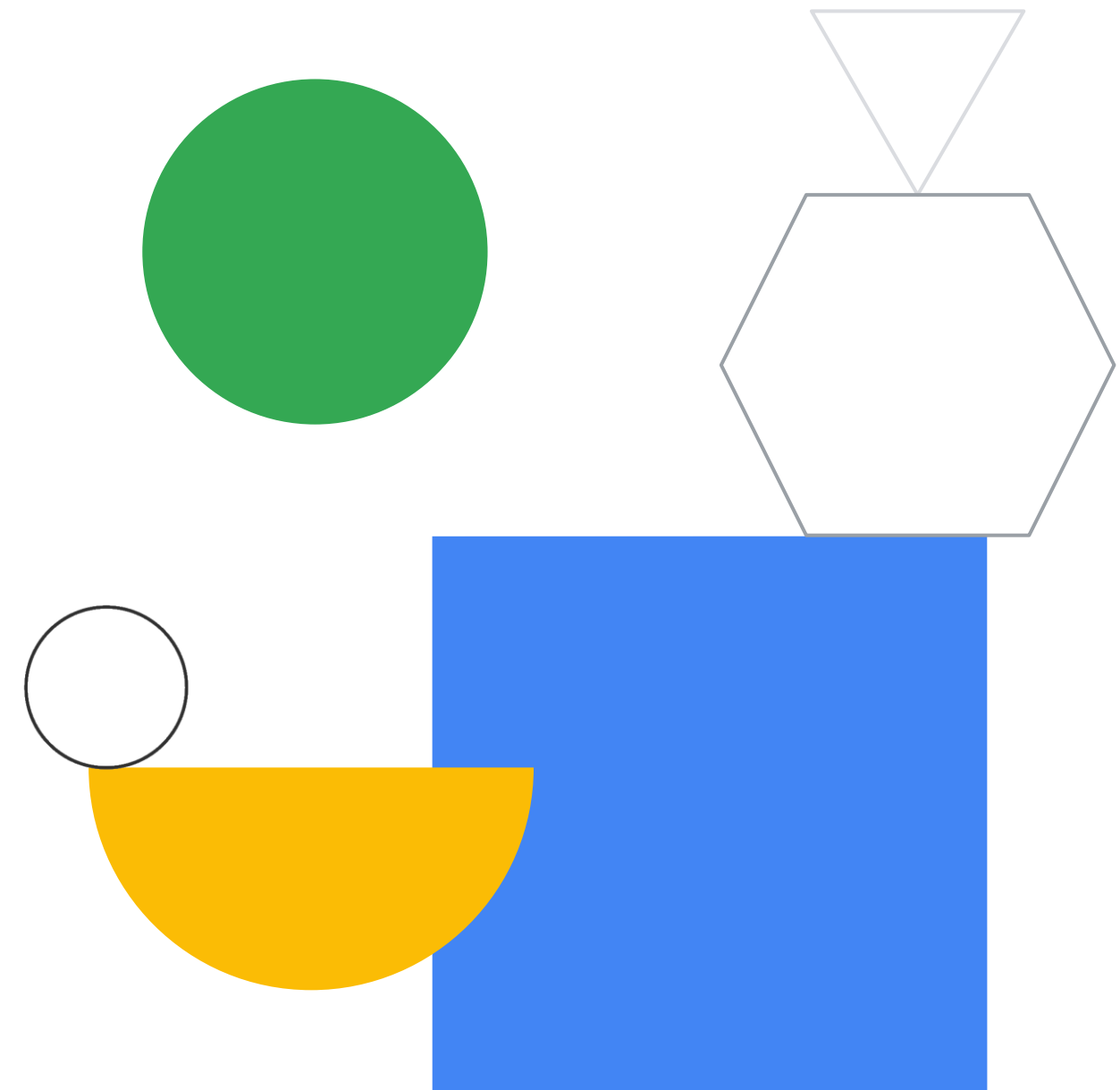
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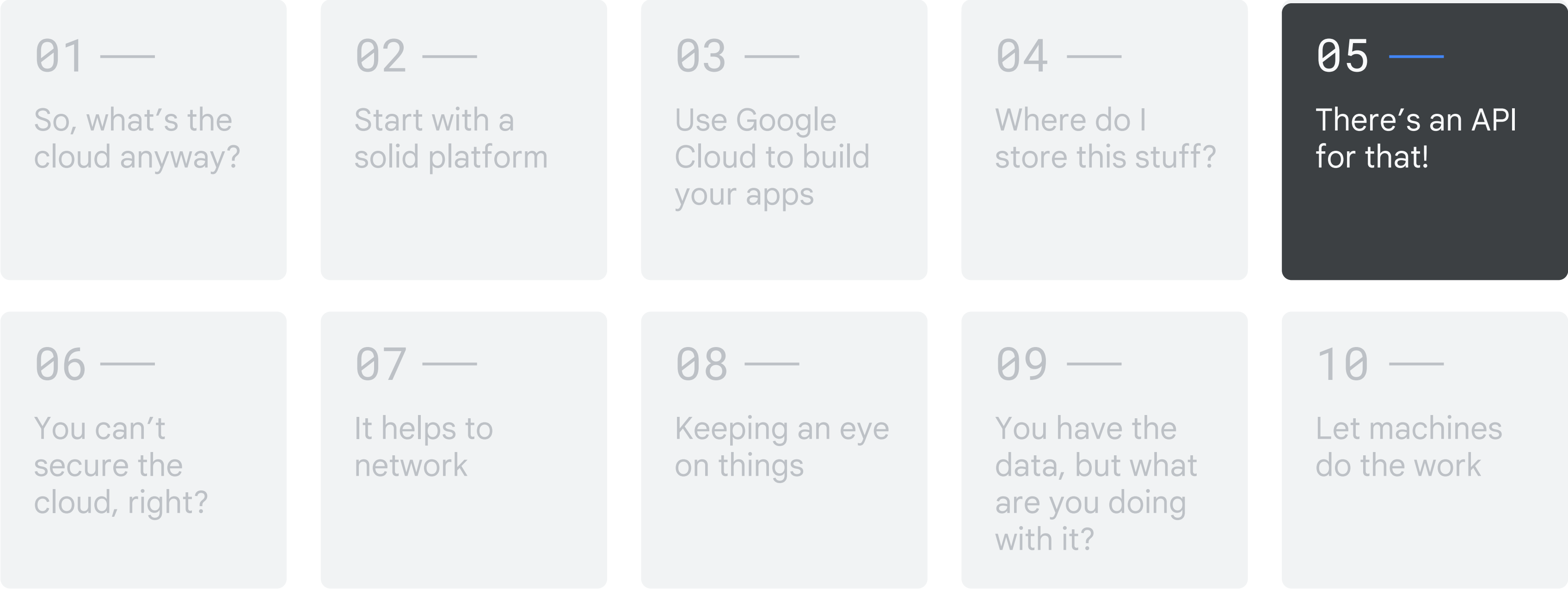


There's an API for that!

Module 5 - APIs
Google Cloud Computing Foundations



Course map



Google Cloud skill badges

Objectives

- 01 Explore the purpose and benefit of APIs.
- 02 Compare Cloud Endpoints and Apigee API Management, two different API management tools.
- 03 Examine Pub/Sub, a Google Cloud tool to handle distributed message-oriented architectures at scale.





Agenda



- 01 The purpose of APIs
- 02 Cloud Endpoints
- 03 Lab: Cloud Endpoints: Qwik Start
- 04 Apigee API Management
- 05 Pub/Sub
- 06 Lab: Pub/Sub: Qwik Start - Python
- 07 Quiz
- 08 Summary

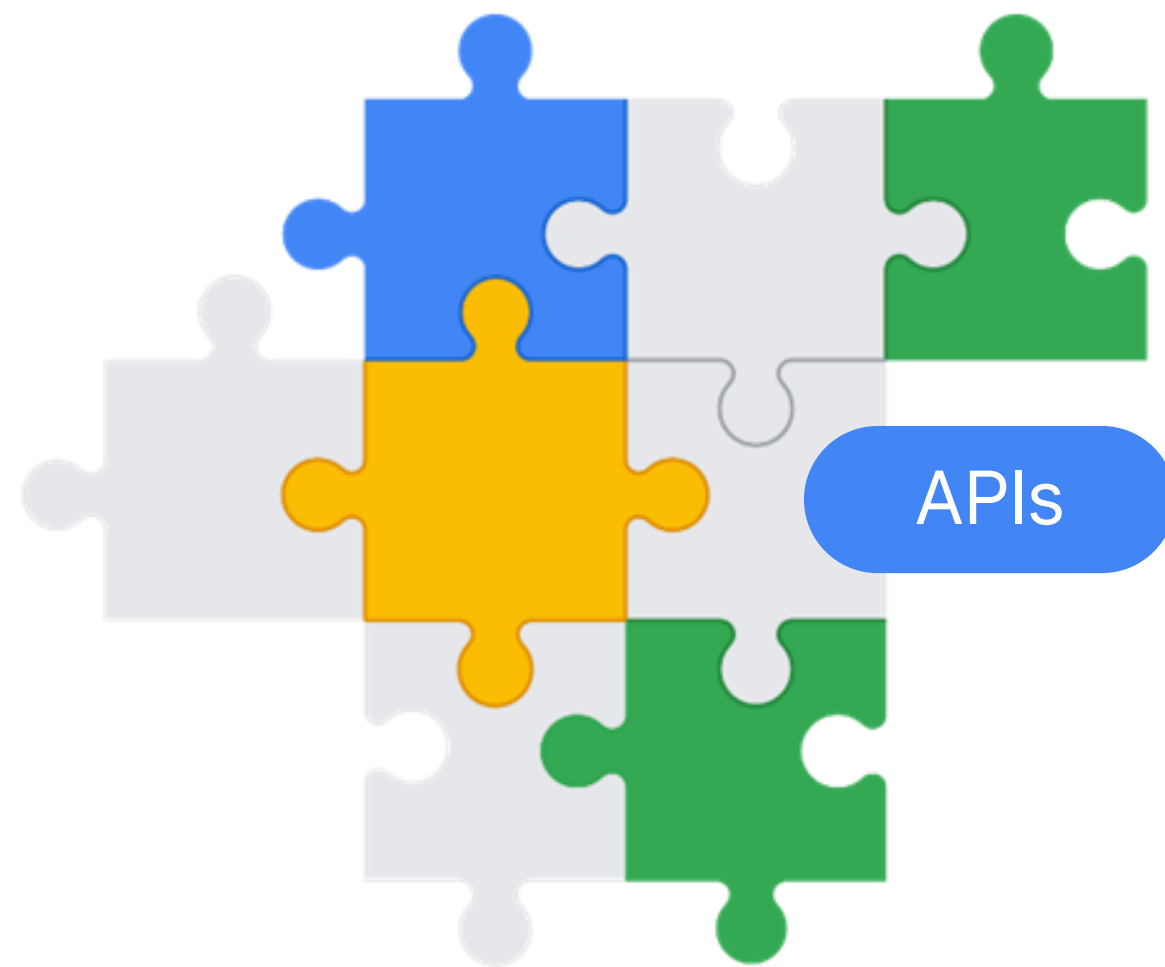


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Application programming interface (API)



- ✓ A clean, well-defined interface
- ✓ Underlying implementation can change
- ✓ Changes to the API are made with versions

What is a REST API?

{REST}

A set of constraints a service must comply with.

An API that uses HTTP requests to GET, PUT, POST, and DELETE data.

Designed to set up a format for applications to communicate.

Great for cloud applications because they are stateless.

Authentication via OAuth and security by leveraging tokens.

Deploying and managing APIs can be difficult



**Interface
definition**



**Authentication
and authorization**



**Logging
and monitoring**



**Management
and scalability**

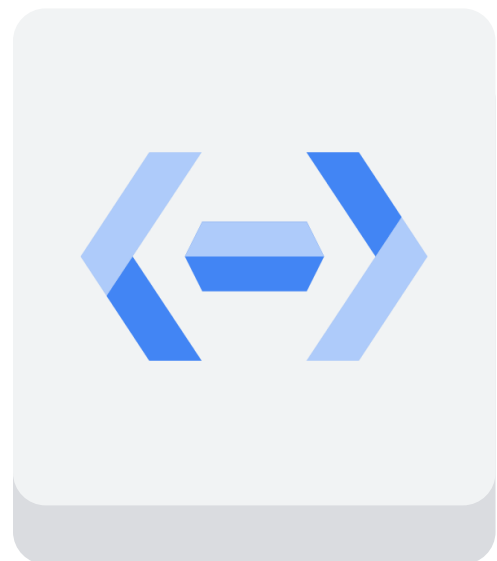


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Cloud Endpoints



Distributed API management system

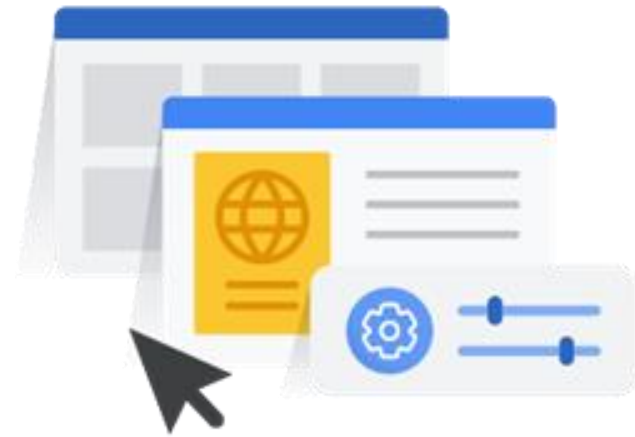
Provides an API Console, hosting, logging, monitoring, and other features

Use with any APIs that support the OpenAPI Specification

Supports applications running in App Engine, Google Kubernetes Engine, and Compute Engine

Clients include Android, iOS, and Javascript

Deploy and manage APIs easily with Cloud Endpoints



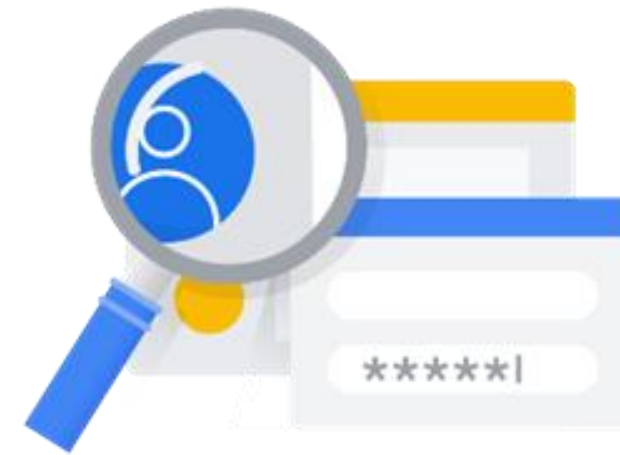
Interface definition

Open API
gRPC API



Authentication and authorization

Service-to-service authentication
User authentication



Logging and monitoring

Cloud Logging
Cloud Trace



Management and scalability

Extensible Service
Proxy Service
Management
Service Control



Agenda



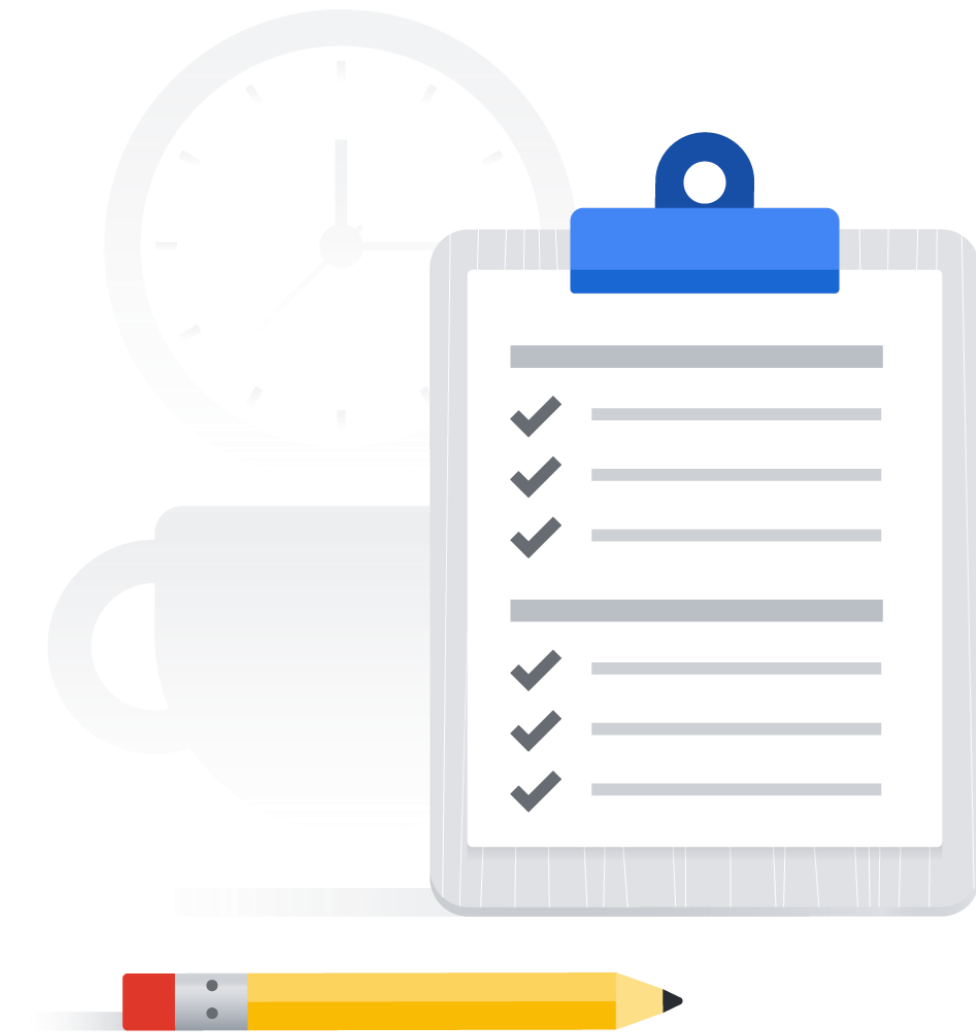
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Cloud Endpoints: Qwik Start

 Hands-on lab

During this lab, you'll:

1. Deploy the Cloud Endpoints configuration.
2. Deploy the API backend.
3. Send requests to the API.
4. Track API activity.
5. Add a quota to the API.





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Apigee API Management



Specific focus on business problems, like rate limiting, quotas, and analytics

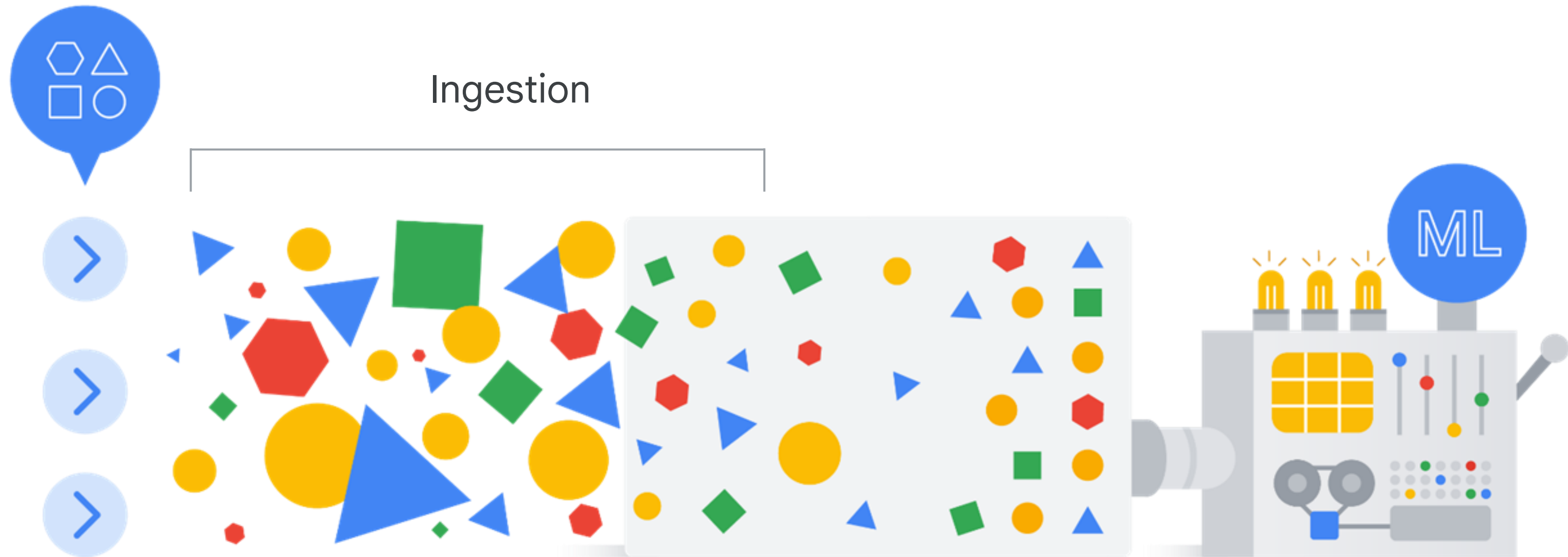
Many Apigee API Management users provide a software service to other companies

Backend services for Apigee API Management don't need to be in Google Cloud

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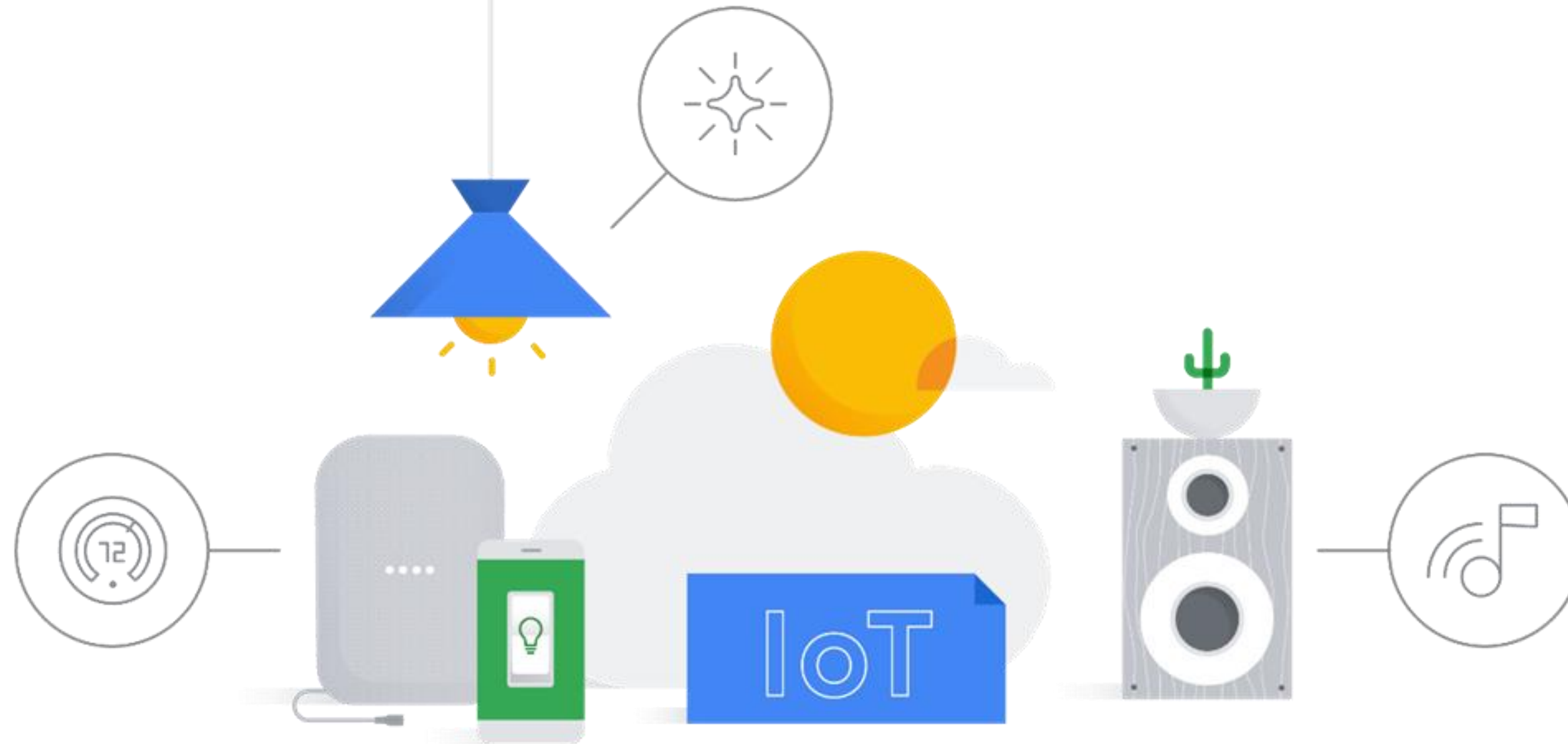
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Data ingestion is when stream data is received



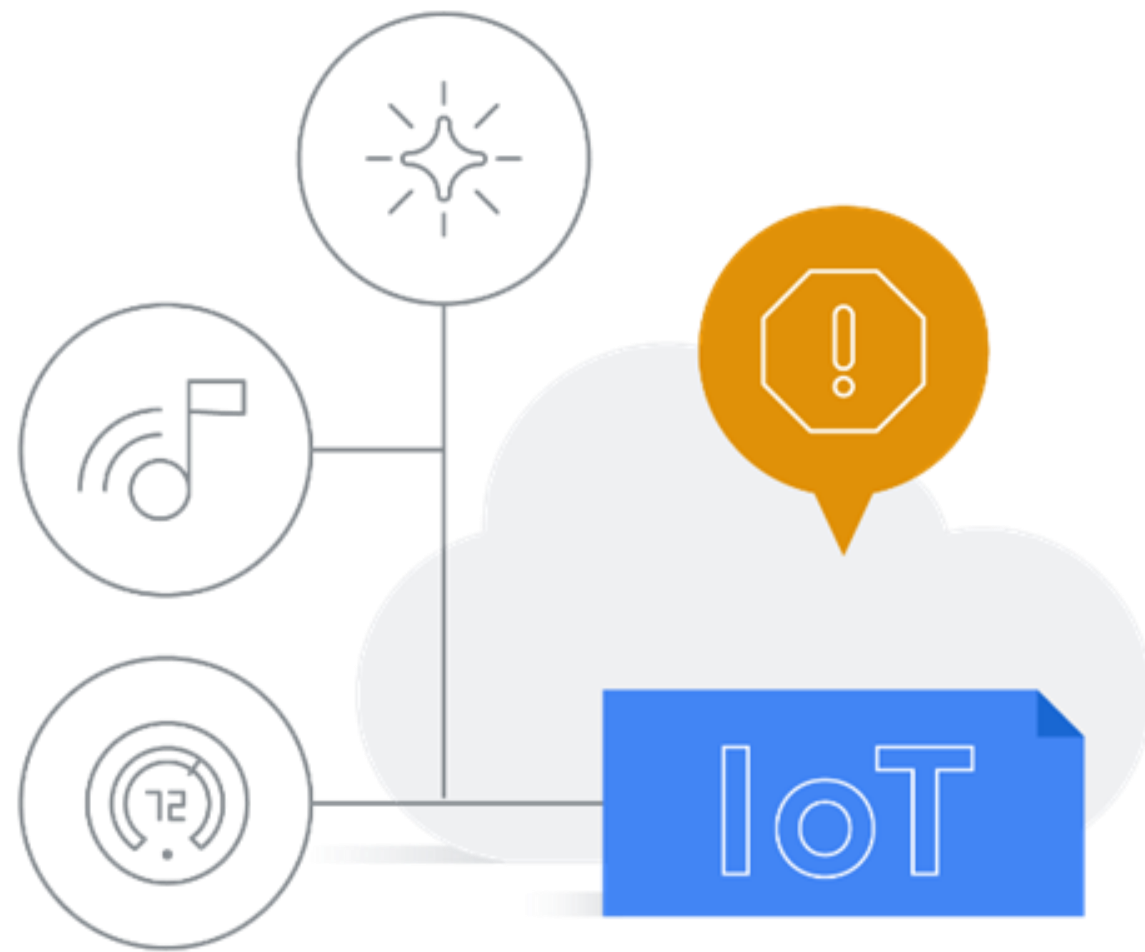
Data pipeline

For example, IoT



Internet of Things applications

Challenges to data ingestion



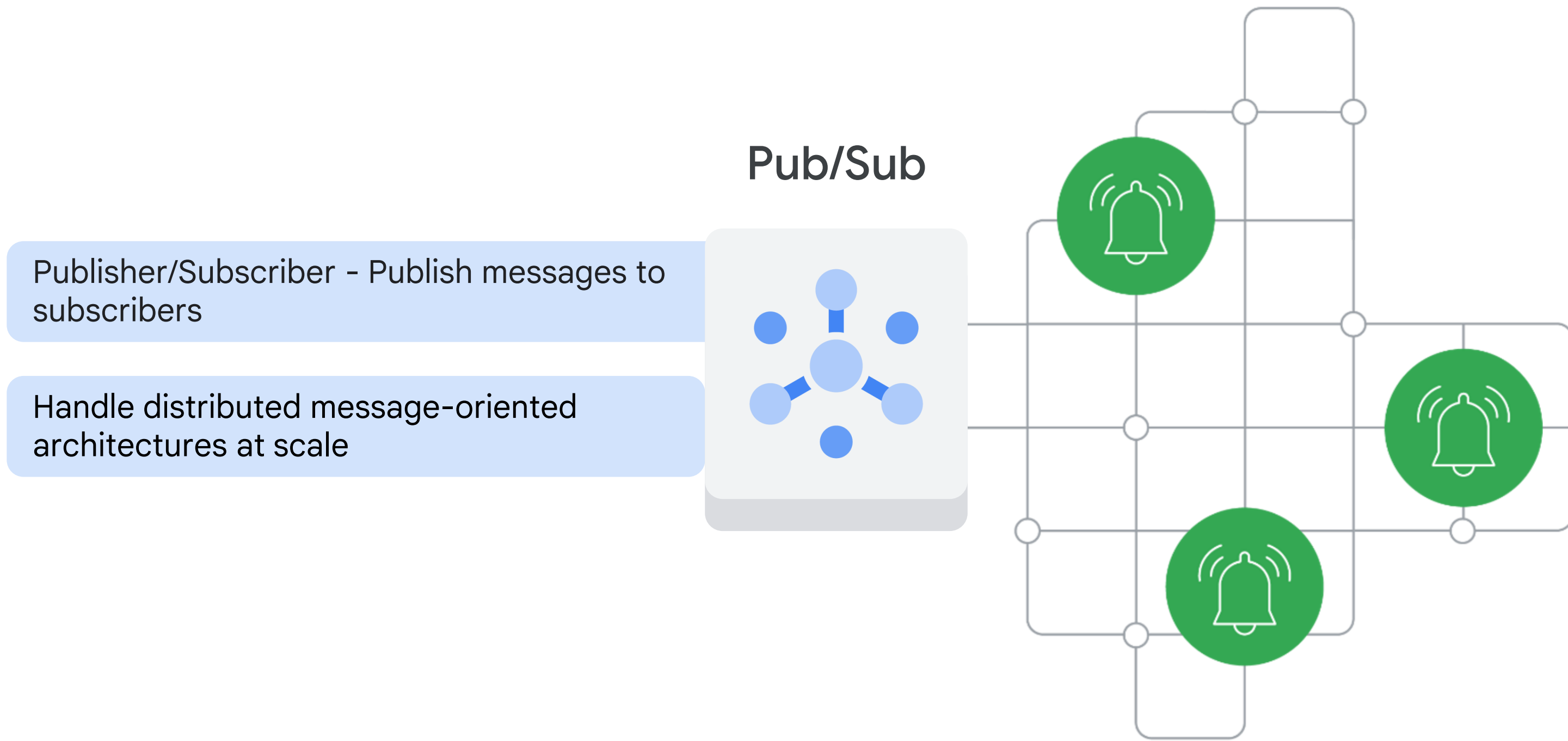
01 Data can be streamed from many different methods and devices

02 It can be hard to distribute event messages to the right subscribers

03 Data can arrive quickly and at high volumes

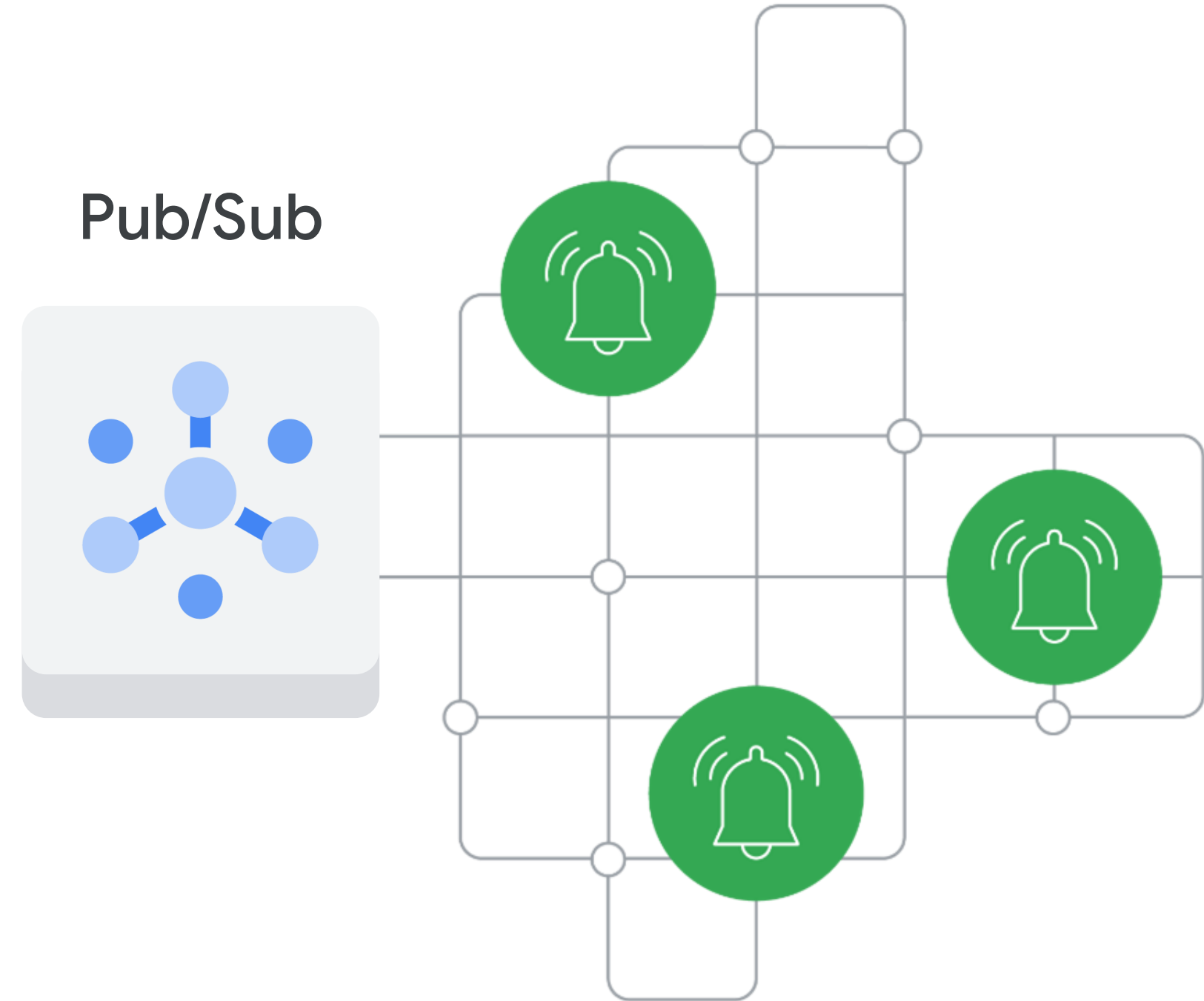
04 Ensuring services are reliable, secure, and perform as expected

Pub/Sub: A distributed messaging service

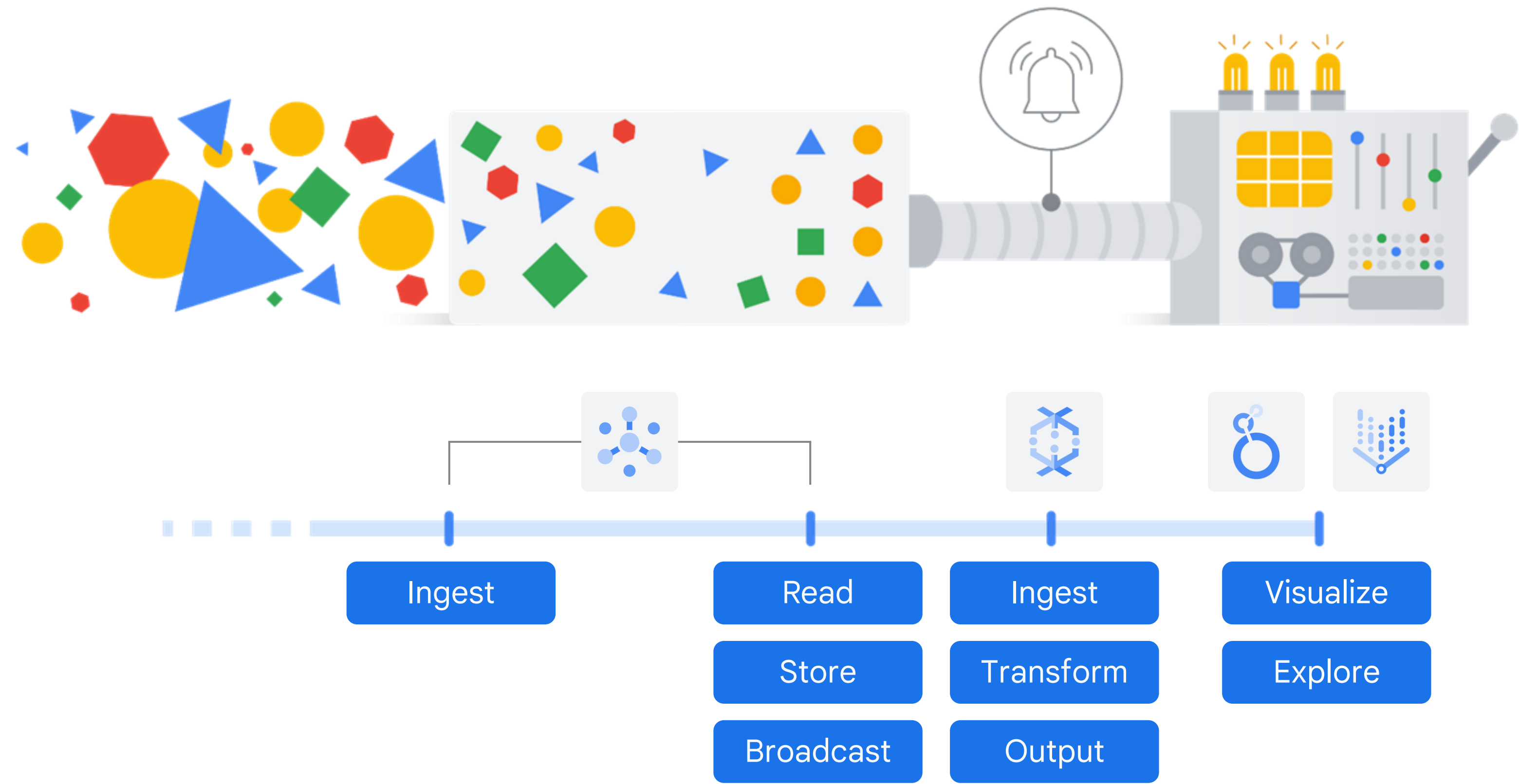


Pub/Sub features

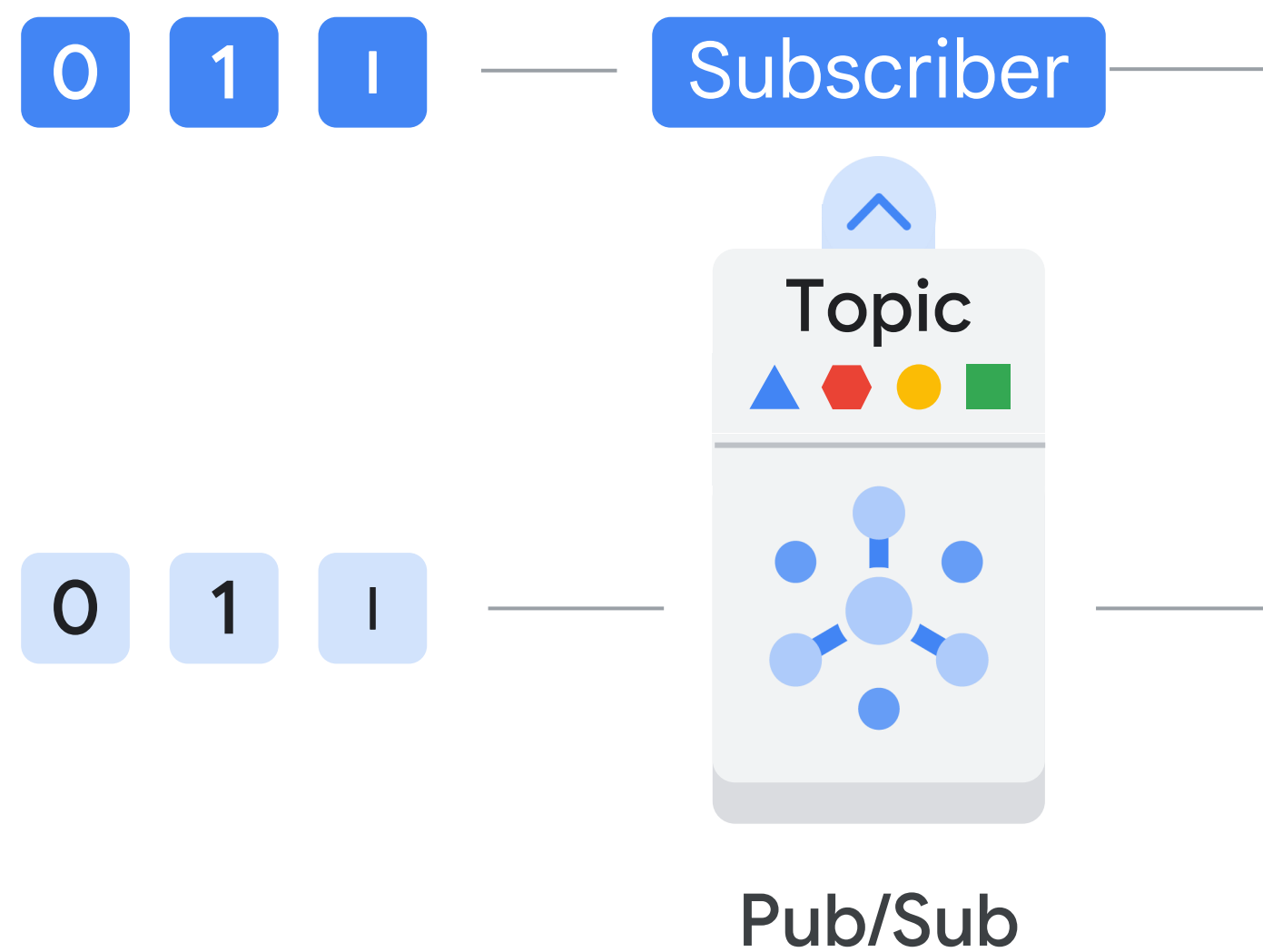
- ✓ Ensures at-least-once delivery
- ✓ No provisioning is required
- ✓ APIs are open
- ✓ Global by default
- ✓ Offers end-to-end encryption



Pub/Sub architecture



Topics



A topic is a central element of Pub/Sub.

Topics and subscribers are decoupled.

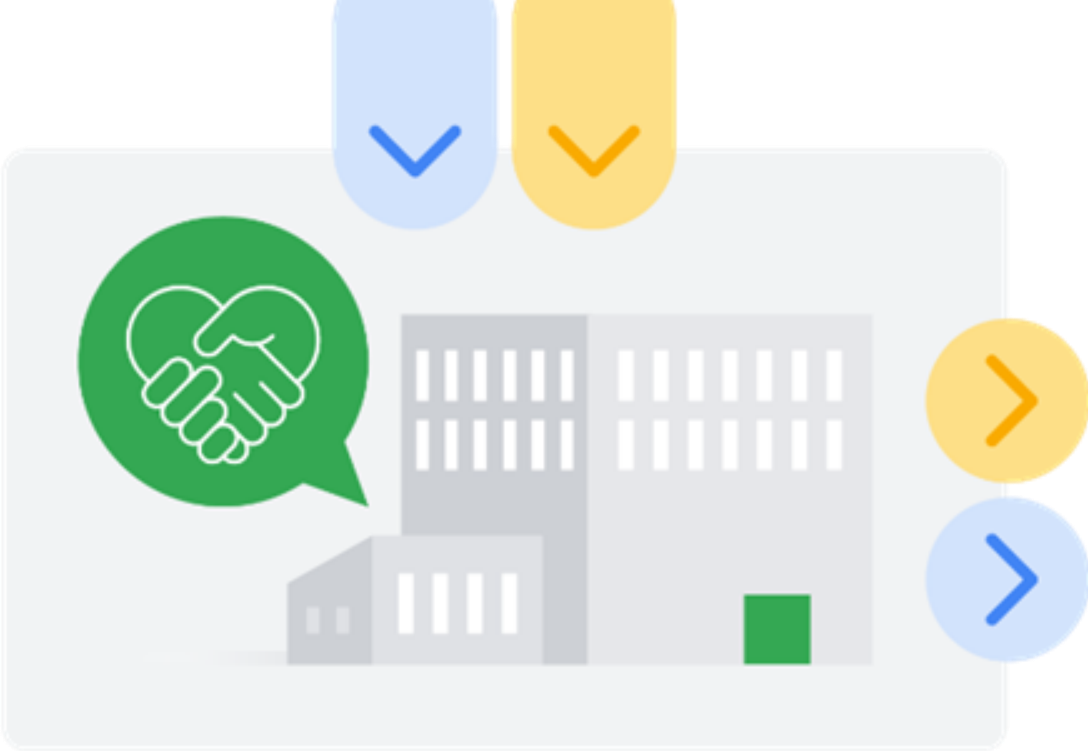
Like a radio antenna, broadcasting no matter the subscribers.

Topics and subscribers can be 0..M to 0..M relationship.

Example: Human resource topics

 New employee 

 New contractor 



Human Resources Topic

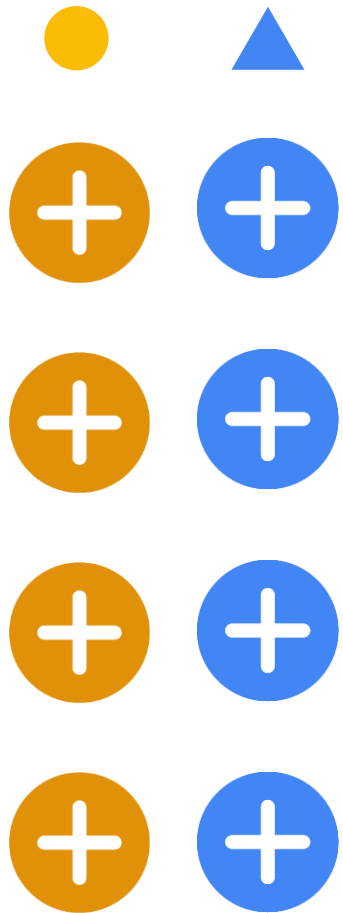


Directory service

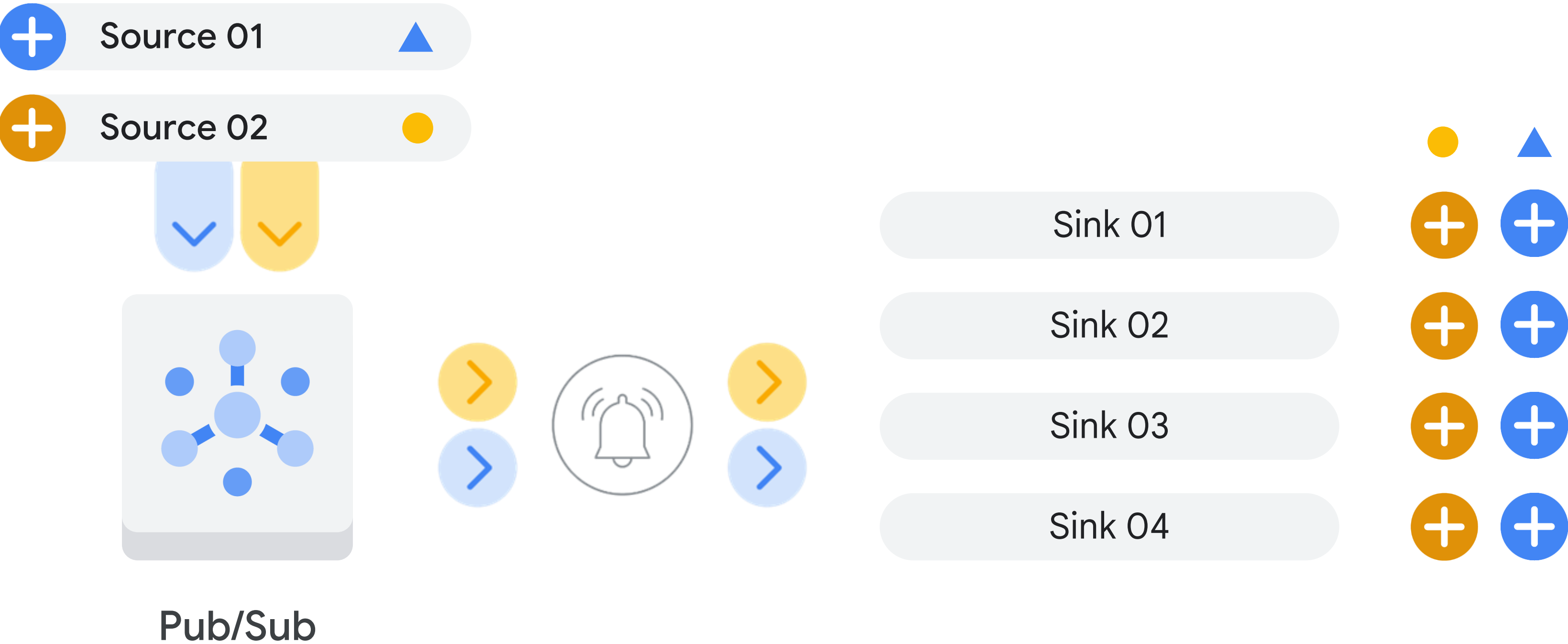
Facilities system

Account provisioning

Badge activation



Pub/Sub ingests streaming data





Agenda



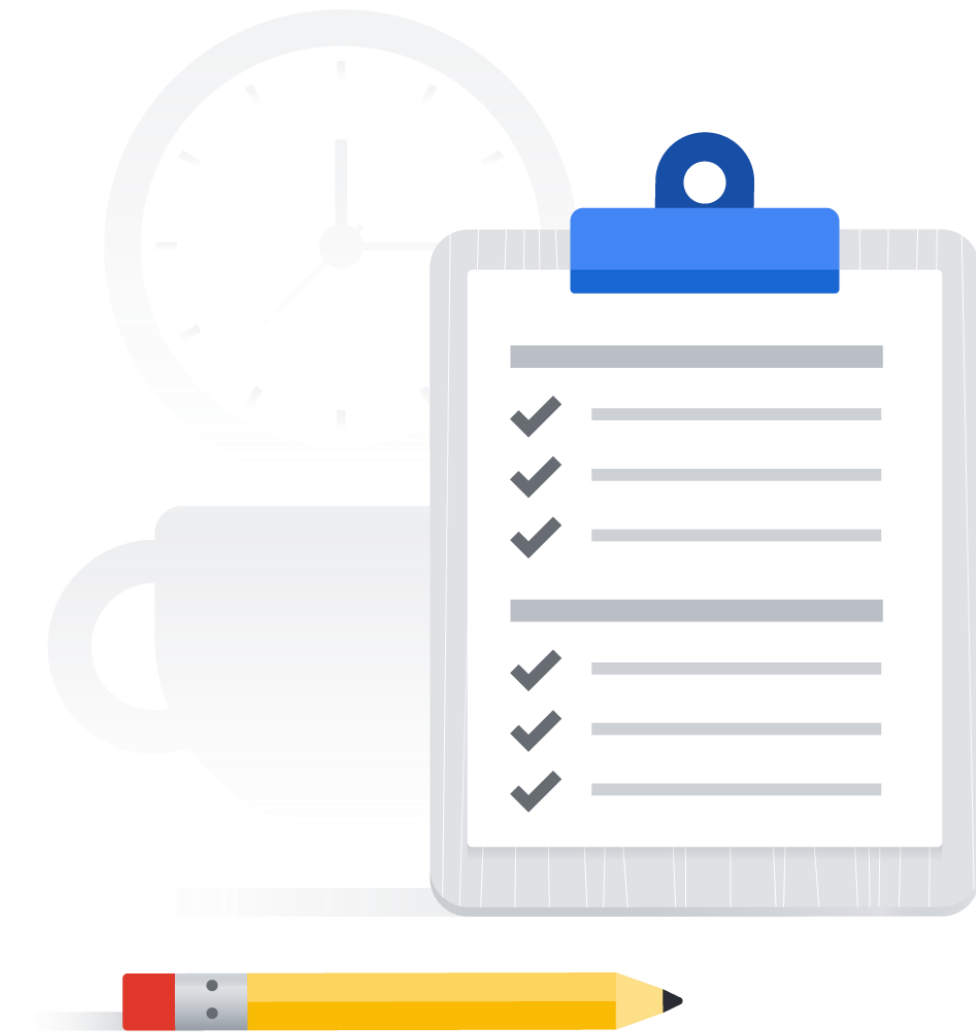
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Pub/Sub: Qwik Start - Python

 Hands-on lab

During this lab, you'll:

1. Learn the basics of Pub/Sub.
2. Create and list a Pub/Sub topic.
3. Create and list a Pub/Sub subscription.
4. Publish messages to a topic.
5. Use a pull subscriber to output individual topic messages.





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Knowledge check

What does API stand for?

- A. Artificial Programming Interface
- B. Application Programming Interface
- C. Asynchronous Programming Interface
- D. Applied Programming Interface



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Knowledge check

Which API management system supports applications running in App Engine, Google Kubernetes Engine, and Compute Engine?

- A. Pub/Sub
- B. Apigee API Management
- C. Cloud Endpoints
- D. REST API



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Knowledge check

Which platform for developing and managing API proxies has a specific focus on business problems, like rate limiting, quotas, and analytics?

- A. Cloud Endpoints
- B. Apigee API Management
- C. REST API
- D. Pub/Sub



Knowledge check

Which platform for developing and managing API proxies has a specific focus on business problems, like rate limiting, quotas, and analytics?

- A. Cloud Endpoints
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Knowledge check

Which statement about Pub/Sub is true?

- A. Pub/Sub is regional by default.
- B. Pub/Sub is unencrypted.
- C. Pub/Sub requires provisioning.
- D. Pub/Sub's APIs are open.



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Summary

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